

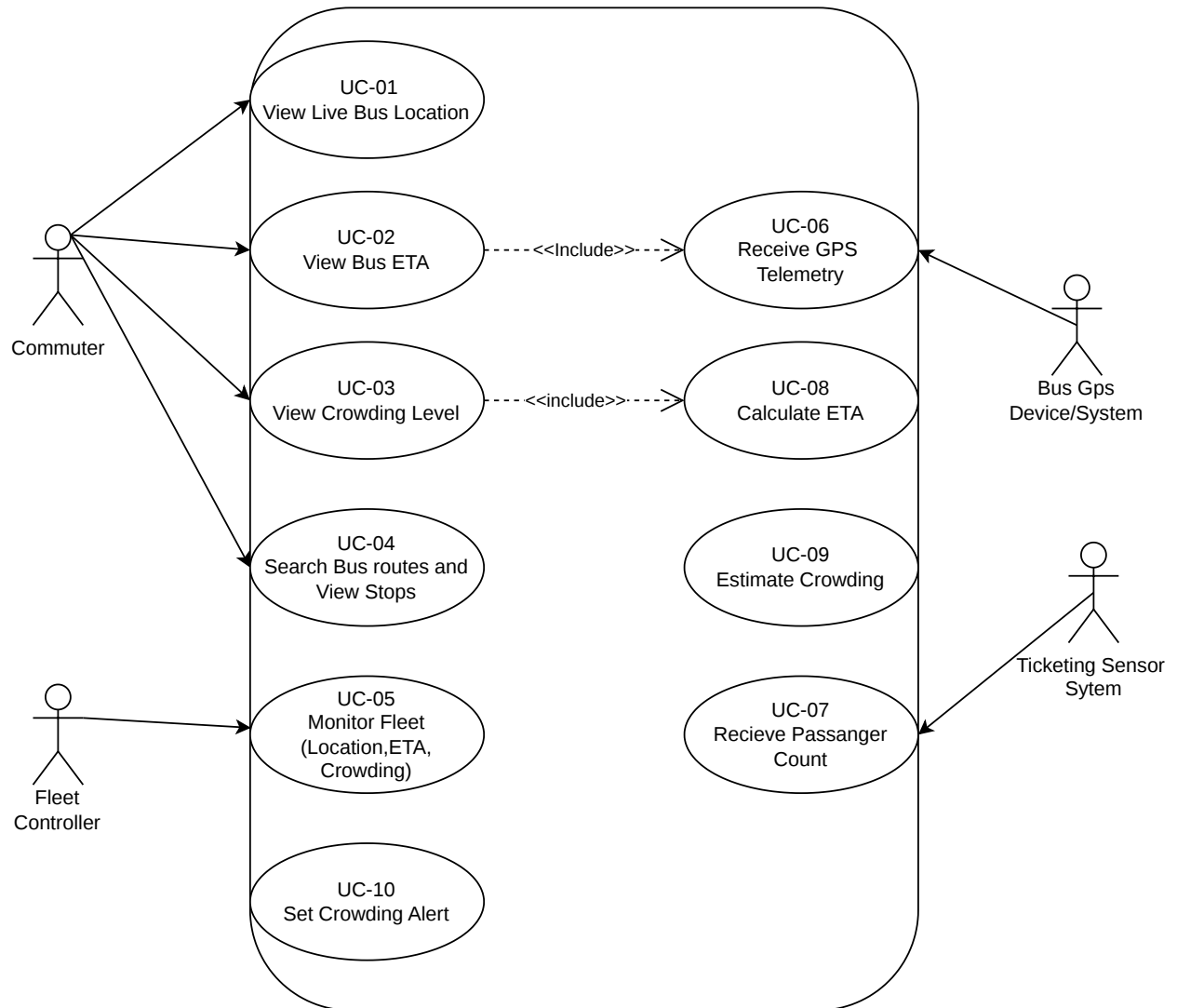
TABLE REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming route stops.	High	Pass: When valid GPS coordinates are received, the system updates the bus position and recalculates ETA for each upcoming stop within 5 seconds. Fail: GPS data is received but the ETA is not updated when the bus position changes.	Real-time GPS tracking and ETA prediction are core functions of the system.
FR-002	Functional	The system shall allow commuters to view the current location of buses operating on a selected route.	High	Pass: After selecting a route, the system displays the current location of active buses on that route and refreshes their positions within 10 seconds. Fail: A bus with valid recent GPS data is not displayed or its location is not updated.	Allows commuters to track the current location of their bus.
FR-003	Functional	The system shall calculate and display the estimated crowding level of each active bus using passenger ticketing sensor data.	High	Pass: When passenger-count data is received, the system calculates and displays a crowding level as Low, Moderate, High, or Full within 10 seconds. Fail: Valid passenger-count data is available but no crowding level is displayed.	Provides commuters and fleet controllers with useful information about passenger load.

TABLE REQUIREMENTS

FR-004	Functional	The system shall allow fleet controllers to monitor the live location, ETA, and crowding level of active buses.	High	Pass: A fleet controller can select an active bus and view its latest GPS location, upcoming-stop ETA, and current crowding level. Fail: Any of these three values cannot be retrieved for a bus with valid telemetry.	Enables fleet controllers to monitor and manage bus operations effectively.
FR-005	Functional	The system shall allow commuters to search for a bus route and view its upcoming stops and estimated arrival times.	Medium	Pass: When a commuter selects a valid route, the system displays the route's upcoming stops and the latest ETA for each stop. Fail: The selected route is displayed without its upcoming stops or ETA information.	Helps commuters plan their journeys using current route and arrival information.
NFR-001	Nonfunctional	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Pass: During load testing with 1,000 active buses transmitting telemetry, the system processes GPS and crowding data with an average processing latency of no more than 2 seconds. Fail: Average telemetry processing latency exceeds 2 seconds or telemetry processing fails for active buses.	Ensures the system remains responsive during peak city-wide transit operations.

Public Bus Live Tracking & Crowding Estimator (UML Diagram)



Use-Case Flow Specification – View Bus ETA

System: Public Bus Live Tracking & Crowding Estimator

Use Case ID: UC-02

Use Case Name: View Bus ETA

Primary Actor: Commuter

Supporting Actor: Bus GPS Device/System

Description

The commuter selects a bus and the system displays its estimated arrival time at upcoming stops based on the latest available GPS data.

Preconditions

1. The commuter has access to the bus tracking platform.
2. The selected bus route exists in the system.
3. Recent GPS telemetry is available for the selected bus.
4. The system is operational.

Postconditions

1. The commuter is shown the estimated arrival times for upcoming stops.
2. The displayed ETA is based on the latest available bus location and telemetry.
3. The system does not display an outdated ETA as current information.

Main Success Scenario

1. The commuter opens the Public Bus Live Tracking & Crowding Estimator.
2. The system displays available bus routes.
3. The commuter selects a bus route.
4. The system displays active buses operating on the selected route.
5. The commuter selects a bus.
6. The system retrieves the latest GPS location and speed of the selected bus.
7. The system identifies the upcoming stops on the selected route.

8. The system calculates the estimated arrival time for each upcoming stop.
9. The system displays the bus's current location and ETA for the upcoming stops.
10. The commuter views the estimated arrival information.
11. The use case ends successfully.

Alternate Flow

6a. GPS Data Unavailable

1. The system attempts to retrieve the latest GPS data.
2. The system determines that valid recent GPS data is unavailable.
3. The system displays a message indicating that the live ETA is temporarily unavailable.
4. The system does not display a misleading or outdated ETA.
5. The commuter may select another active bus or try again later.
6. The use case ends.